Sebastian A. J. Lortz

Curriculum Vitae

PhD Candidate in Psychometrics, University of Groningen
s.a.j.lortz@gmail.com · ORCID · GitHub · LinkedIn

Research Profile

In my PhD project, *Models for Computer Adaptive Testing of Cognitive Functions* (2025–2029, funded by KWF), I develop the psychometric models and algorithms behind a home-based computerized adaptive test (CAT) that screens cancer survivors for cognitive impairment. The project is a collaboration with the Netherlands Cancer Institute and the software provider Eyra. My work focuses on:

- developing item response theory models for neuropsychological test data;
- modelling response accuracy and response time jointly to improve inference;
- calibrating item banks with these models efficiently;
- developing item selection algorithms for computerized testing;
- combining sub-test outcomes into a single actionable decision.

Supervisors: Dr. Muirne C. S. Paap and Dr. Niek Frans (University of Groningen); Prof. Sanne B. Schagen and Dr. Joost A. Agelink van Rentergem (Netherlands Cancer Institute).

Positions

PhD Candidate in Psychometrics, University of Groningen | 2025, September–present
Models for Computer Adaptive Testing of Cognitive Functions, funded by KWF.

Research Assistant, University of Groningen | 2025, January–2025, July
Re-analysed cross-national studies with spatial and phylogenetic Bayesian regression methods to account for cross-national statistical dependencies.

Research Assistant, University of Groningen | 2024, February–2025, February
Developed two new sample-size planning methods for Bayesian interval testing in 2 × 2 factorial designs, integrated them with two existing methods into an R package and a server-hosted Shiny app, and coordinated the project with (inter)national collaborators.

Statistics and Research Consultant, The Methodology Shop, University of Groningen | 2024, January–2025, August
Delivered statistical consultation and methodological guidance to (Ph.D.) students and staff, covering Bachelor/Master theses, Ph.D. dissertations, (inter)national research projects, and proposals for theses and doctoral studies.

Research Assistant, University of Groningen | 2023, October–2023, December
Collected data for an experimental study and monitored the study's progress.

Teaching Assistant, Statistics II, B.Sc. Psychology, University of Groningen | 2022, September–2025, January
Prepared and facilitated practical sessions on statistical methods in the Statistics II course.

Education

M.Sc. Research Behavioural and Social Sciences, Specialisation in Psychometrics & Statistics, University of Groningen | September 2023–July 2025, *cum laude*.
- **Elective Courses:** Structural Equation Modelling; Multilevel Analysis; Network Analysis; Statistical Modelling of Single Cases; Complexity, Dynamics and Development; Modelling Interactions of Persons and Variables over Time.
- **Projects:** From Problem Analysis to Intervention Design; Multidisciplinary Research in Action; Applied Statistics.
- **Master Thesis:** Advancing Replication Study-Decision Making with DISCOURSE: Data-Simulation via Iterative Stochastic Combinatorial Optimization Using Reported Summary Estimates.

B.Sc. Psychology, Free Choice Minor, University of Groningen | September 2020–August 2023.

B.Sc. Health Sciences, Free Choice Minor, Technical University of Munich | October 2016–August 2020.
- **ERASMUS:** Semester at the University of Vienna, Faculty of Psychology.

Publications

Journal Articles

Lortz, S. A. J., Setiono, A., Kovacs, M., & van Ravenzwaaij, D. (2026). Sample-size planning for frequentist and Bayesian 2 × 2 analysis-of-variance designs. *Advances in Methods and Practices in Psychological Science, 9*(1). https://doi.org/10.1177/25152459251398805

Paap, M. C. S., **Lortz, S. A. J.**, Hummelen, B., Fjermestad-Noll, J., Kvarstein, E., & Pedersen, G. (2025). An investigation of longitudinal measurement invariance and item response theory analysis of the LPFS-BF 2.0 in a large clinical Norwegian sample. *Personality and Mental Health, 19*(3), e70032. https://doi.org/10.1002/pmh.70032

Submitted

Lortz, S. A. J., Frans, N., Veldkamp, B. P., Agelink van Rentergem, J. A., Schagen, S. B., & Paap, M. C. S. (2026). Sample-Size Planning for Item Calibration with the Joint Hierarchical Model of Response Accuracy and Response Time. Manuscript submitted for publication.

Lortz, S. A. J., Post, W. J., Koopman, L., & Paap, M. C. S. (2026). Stability and Scalability in Mokken Scale Analysis in the Presence of Small Samples and Deviant Items with Dichotomous Response Data. Manuscript submitted for publication.

Lortz, S. A. J., Timmerman, M. E., & van Ravenzwaaij, D. (2026). Simulating Raw Data from Reported Summary Statistics: A Non-parametric Approach. Manuscript submitted for publication.

Software

Lortz, S. A. J., Setiono, A., Kovacs, M., & van Ravenzwaaij, D. (2025). Sample Size Planning for Frequentist and Bayesian 2 × 2 ANOVA Designs (v0.2.0). *Zenodo.* https://doi.org/10.5281/zenodo.17362197

Presentations

Lortz, S. A. J., Frans, N., Veldkamp, B. P., Agelink van Rentergem, J. A., Schagen, S. B., & Paap, M. C. S. (2026, November). Sample-Size Planning for Item Calibration with the Joint Hierarchical Model of Response Accuracy and Response Time [Conference Presentation]. 40th IRT Workshop.

Lortz, S. A. J. (2026, November). Introduction to R for Staff Members [Workshop]. University of Groningen, Groningen, Netherlands.

Lortz, S. A. J., Post, W. J., Koopman, L., & Paap, M. C. S. (2024, November). Stability and Scalability in Mokken Scale Analysis: Challenges with Small Samples and Deviant Items [Conference Presentation]. 38th IRT Workshop, University of Twente, Enschede, Netherlands. Slides

Lortz, S. A. J. (2024, April). A Simple Introduction to Monte Carlo Simulations for a Priori Power Analyses [Seminar Presentation]. University of Groningen, Groningen, Netherlands. Slides R Code

Grants & Awards

Cum laude, M.Sc. Research Behavioural and Social Sciences, University of Groningen | 2025, July

Teaching

Workshops

Introduction to R for Staff, Nieuwenhuis Institute, University of Groningen, Groningen | 2026–2029
Workshops introducing university staff members to data analysis in R.
Materials: Material

Introduction to R for Students, Pedagogical Sciences, University of Groningen, Groningen | 2026–2027
Developed six weeks of practical R materials that students work through in supervised sessions.
Materials: Material

Supervision

Master Thesis Supervision, Pedagogical Sciences, University of Groningen, Groningen | 2026–2027
Supervising a Master's thesis on piloting computer-based tests of cognitive functioning and analyzing the resulting data.

Bachelor Supervision, Pedagogical Sciences, University of Groningen, Groningen | 2025–2026
Supervised a group of Bachelor students on parametric and nonparametric Item Response Theory (IRT) analyses, guiding their methodological choices, data analyses, and interpretation of results.

Courses

IRT Workshop, Pedagogical Sciences, University of Groningen, Groningen | 2025–present
Taught practical sessions on the application of parametric and nonparametric Item Response Theory (IRT) methods.

Test Theory, Pedagogical Sciences, University of Groningen, Groningen | 2025–present
Taught practical sessions on test theory, using datasets generated for the exercises.

Applied Statistics, Research Master Behavioural Sciences, University of Groningen, Groningen | 2025–present
Taught seminars on the application of statistical methods in R. Supported students in analyzing their own research projects with a variety of techniques.

SPSS Practical, Pedagogical Sciences, University of Groningen, Groningen | 2025–2026
Supervised practical sessions on conducting statistical analyses in SPSS.

Research Methods and Techniques, Teacher Education, University of Groningen, Groningen | 2025–2026
Supervised practical sessions.

Statistics II, Psychology, University of Groningen, Groningen | 2022–2025
Taught seminars on the theory and application of basic statistical methods, including multiple linear regression, mixed ANOVA, and Bayesian statistics, using JASP.

Consulting

Methodological Consulting, Pedagogical Sciences, University of Groningen, Groningen | 2026–present
Methodological and statistical advice for researchers and students in the department.

Statistics Tutor, Freelance, Groningen, Groningen | 2020-2025
Provided individualized tutoring to enhance students' comprehension and application of statistical analyses, along with tailored methodological consulting for thesis projects.

Professional Development

Interuniversity Graduate School of Psychometrics and Sociometrics (IOPS)

Optimization and Numerical Methods, Geert Molenberghs, Francis Tuerlinckx, Katrijn Van Deun & Tom Wilderjans, KU Leuven | 2026, December (2 ECTS)
Iterative optimization algorithms for estimating statistical model parameters, their connections to each other and to statistical inference, and related numerical methods such as numerical differentiation and integration, implemented in R.

The Use of Data in Decision-Making, Including Standardized Psychological Tests, Marieke Timmerman & Rob Meijer, University of Groningen | 2026, November (2 ECTS)
Norm- and criterion-referenced interpretation of standardized test scores, how norms are constructed and reported, and how multiple pieces of information are combined in decisions, including common flaws of holistic judgment and structured alternatives that improve decision quality.

Bayesian Item Response Theory, Jean-Paul Fox, University of Twente | 2026, October (2 ECTS)
Bayesian IRT modelling and MCMC estimation in R (JAGS, Stan), with applications to small samples and differential item functioning.

What is Psychometrics?, Dylan Molenaar (coordinator), University of Amsterdam | 2026, September (2 ECTS)
Overview of psychometrics as a research field, covering measurement in network models, response-time modelling, values in psychometrics, theory-based hypothesis evaluation with GORIC(A), Bayesian graphical modelling, and the within/between problem.

Simulation-Based Inference for Cognitive Modeling, Paul-Christian Bürkner & Stefan Radev | 2026, April (1.6 ECTS)
Hands-on introduction to simulation-based (amortized Bayesian) inference with the BayesFlow ecosystem, from simulation to model deployment.

Statistical Learning and Prediction, Tom Wilderjans & Marjolein Fokkema, Leiden University | 2026, January (2 ECTS)
Provided a comprehensive introduction to statistical learning techniques, including supervised and unsupervised methods, based on the book *Introduction to Statistical Learning* https://www.statlearning.com. Focused on building theoretical understanding and practical skills for applying statistical learning methods in empirical research.

Mathematics Courses

Calculus 2 (Multivariable Calculus), B.Sc. Computing Science, University of Groningen | 2026, October (5 ECTS)
Partial and directional derivatives, constrained optimization with Lagrange multipliers, multiple integrals, and Fourier series.

Linear Algebra, B.Sc. Computing Science, University of Groningen | 2026, June (5 ECTS)
Systems of linear equations, vector spaces, linear transformations, determinants, eigenvalues and eigenvectors, and orthogonality.

Teaching & Supervision Training

Start to Teach, University of Groningen | 2026, February (1.5 ECTS)
Workshop series on planning and preparing teaching activities, aligning them with learning outcomes, activating students, creating a safe learning environment, and giving feedback.

Start to Supervise, University of Groningen | 2026, February (0.4 ECTS)
Training for daily supervision of Bachelor and Master theses: the supervisor's role, organizing the supervision process along the thesis timeline, effective communication, and formative feedback.

Seminars & Workshops

University of Groningen: Markov Chain Monte Carlo Sampling, Socio–Semantic Networks, Generative AI Workshop, Cluster Analysis and Mixture Models, Causality in Observational Data, Mixture Modelling of Longitudinal Data, Regression for Categorical Data, Thematic Analysis, and Systematic Review and Meta-Analysis.

Future Learn: Privacy in Research, General Data Protection Regulation, and Data Protection Agreements.

Service & Memberships

Member, PhD Selection Committee, University of Groningen | 2026

Member, Interuniversity Graduate School of Psychometrics and Sociometrics (IOPS) | 2025–present

Skills

Statistical Modelling:
- **Latent Variable Methods:** (Non-)parametric Item Response Theory, Mokken Scaling, Response Time Models, Structural Equation Modelling.
- **Supervised Methods:** Regression Models (Generalized, Penalized, Spatial, Splines, GAM), Regression Trees (bagging, random forest, boosting, BART), Network Analysis.
- **Unsupervised Methods:** K-means, Hierarchical Clustering, Principal Components.
- **Time-Series & Other Analyses:** (Non-linear) Time-Series Models, Cross-Recurrence Quantification Analysis, Detrended Fluctuation Analysis.

Algorithms: Markov Chain Monte Carlo, Gibbs Sampling, Simulated Annealing, Particle Swarm Optimization, Hill Climbing.

Languages: English (C1–C2), German (Native), Dutch (A2).

Software & Services: R, GitHub, Quarto, LaTeX, SPSS, JASP.